

IS 302-2-24 : 1994

(Superseding IS 10542 : 1983)

(Reaffirmed 2014)

भारतीय मानक

घरेलू और सम्बद्ध विद्युत साधनों की सुरक्षा

भाग 2 विशेष अपेक्षाएँ

अनुभाग 24 रेफ्रिजरेटर, खाद्य प्रशीतित्र तथा बर्फ तैयार करने वाले साधन

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 24 Refrigerators, Food-Freezers and Ice-Makers

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Price Group 8

Electrical Appliances Sectional Committee, ET 32

FOREWORD

This Indian Standard was adopted by the Bureau of Indian Standards, after the draft finalized by the Electrical Appliances Sectional Committee had been approved by the Electrotechnical Division Council.

This standard covers general and safety requirements of refrigerators, food-freezers and ice-makers to ensure personal safety against electric shock, safety against the effects of excessive temperature and fire. This standard does not cover the performance requirement which are covered under separate Indian Standards.

This standard is to be read in conjunction with IS 302-1 (1979). For the sake of convenience, the clauses of this standard correspond to those of IS 302-1 (1979). Clauses which are applicable (which means that relevant provisions of the clauses apply) or not applicable and necessary changes wherever required and indicated accordingly. Clauses/sub-clauses/tables which are additional to those of IS 302-1 (1979) are numbered starting from 101; additional Annexes are lettered AA. Should, however, any deviation exist between IS 302-1 (1979) and this standard, the provisions of the latter shall apply.

While preparing this standard, assistance has been derived from IEC Pub 335-2-24 (1992) 'Safety of household and similar electrical appliances: Part 2 Particular requirements for refrigerators, food freezers and ice-maker, Section 24 Refrigerators, food-freezers and ice-makers' issued by the International Electrotechnical Commission (IEC).

This standard when published will supersede IS 10542 : 1983 'Specification for safety requirements for refrigerator and food freezer for household and similar purposes', the requirements of which have been covered in this revised standard.

The Indian Standards which are necessary adjuncts to this standard are given in Annex CC.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (revised)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

SAFETY OF HOUSEHOLD AND SIMILAR ELECTRICAL APPLIANCES

PART 2 PARTICULAR REQUIREMENTS

Section 24 Refrigerators, Food-Freezers and Ice-Makers

1 SCOPE

This clause of IS 302-1 (1979) is applicable except as follows.

1.1 Replacement

This standard applies to:

- Electrical refrigerators with or without frozen food storage compartment, frozen food storage cabinets and food freezers and their combinations for household and similar use;
- Ice-makers incorporating a motor-compressor and ice-makers intended to be incorporated in food freezers or frozen food storage compartments;
- Refrigerators and freezers for camping use, touring caravans and for boats for leisure purposes. These appliances may be mains operated, battery operated or operated either by mains or battery. They may also be in addition energized by another source (gas or fuel).

It deals with safety for such appliances. It does not cover features of construction and operation of refrigerators and freezers which are covered under separate Indian Standards.

Some tests related to stability and mechanical hazards are given in Annex BB.

Where applicable, clauses of IS 302-1 (1979) related to safety of heating appliance apply to absorption-type refrigerators or food freezers, and to heating elements incorporated in compression-type refrigerators or food freezers.

Appliances not intended for normal household use, but which nevertheless may be a source of danger to the public, such as appliances intended to be used by laymen in shops, in light industry and on farms, are not within the scope of this standard.

This standard does not take into account the special hazards which exist in nurseries and other places where there are young children or aged or infirm persons without supervision; in

such cases additional requirements may be necessary.

This standard does not apply to:

- Appliances designed exclusively for industrial purposes;
- Appliances intended to be used in locations where special conditions prevail, such as the presence of a corrosive or explosive atmosphere (dust, vapour or gas); and
- Separate motors.

For appliances intended to be used in vehicles or on boardships or aircraft, additional requirements may be necessary.

2 TERMINOLOGY

This clause of IS 302-1 (1979) is applicable except as follows.

2.30 Addition

Normal load for ice-maker — Load obtained when the ice-maker is operated at the room temperature of $22 \pm 3^\circ\text{C}$ and with any controls adjusted to their most unfortunate setting if the setting can be altered by the user.

Normal load for incorporated ice-maker — Load obtained when the ice-maker is operated at the temperature of the frozen food storage compartment as prescribed by the relevant subclauses of this standard, with the supply water at a temperature of $22 \pm 2^\circ\text{C}$ and with any control adjusted to their most unfavourable setting if the setting can be altered by the user.

Additional Subclauses

2.101 Refrigerator

Enclosed thermally insulated appliance of suitable volume and equipment for household use, cooled by a device consuming electric energy and having one or more compartment intended for the preservation of foodstuffs, one of them at least being maintained at a temperature above 0°C .

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2.102 Frozen Food Storage Compartment

Enclosed thermally insulated compartment of a refrigerator whose temperature is such that under the ambient reference conditions foodstuffs placed in it at a temperature of -6°C , -12°C or -18°C are maintained at this temperature or below.

2.103 Frozen Food Storage Cabinet

Enclosed thermally insulated appliance or cabinet suitable for household use, cooled by means of a device consuming electric energy and having one or more compartments whose temperature is such that under ambient reference conditions foodstuffs placed therein, are maintained at a temperature equal to or below -18°C .

2.104 Food Freezer

Enclosed thermally insulated appliance or cabinet of suitable volume and equipment for household use, cooled by means of a device consuming electric energy, and furnished with one or more compartments intended for the freezing of foodstuffs and suitable for the preservation of such foodstuffs at a temperature equal to or below -18°C .

2.105 Compression-Type Appliance

Appliance in which refrigeration is effected by the vaporization at low pressure in a heat exchanger (evaporator) of a liquid refrigerant, the vapour thus formed being restored to the liquid state by mechanical compression to a higher pressure and subsequent cooling in another heat exchanger (condenser).

2.106 Ice-Maker

Appliance in which ice is made by freezing water by a device consuming electric energy and having a compartment of storing the ice.

2.107 Incorporated Increment

Ice-maker specially designed to be incorporated into food freezer or a frozen food storage compartment and without means for freezing water.

2.108 Ancillary Heating Element

Heating device such as a defrost heater, door heater, anti-condensation heater, etc, which performs an auxiliary function.

2.109 Absorption-Type Appliance

Appliance in which refrigeration is effected by the evaporation in a heat exchanger (evaporator) of a liquid refrigerant, in the liquid state, the resulting vapour being then absorbed by an absorbent medium from which

it is subsequently expelled at a higher partial vapour pressure by heating and liquefied by cooling in another heat exchanger (condenser).

2.110 Condenser

Heat exchanger in which after compression vaporized refrigerant is liquefied by rejecting heat to an external cooling medium.

2.111 Evaporator

Heat exchanger in which after pressure reduction, the liquid refrigerant is vaporized by absorbing heat from the medium to be refrigerated.

2.112 Free-Standing Appliance

Appliance intended for open-type installation only.

2.113 Steady-State

Considered to be obtained when three successive readings of the temperature, taken at approximately 60 min intervals, at the same point of any operating cycle, do not differ by more than 1°C .

3 GENERAL REQUIREMENTS

The clause of 302-1 (1979) is applicable.

4 GENERAL NOTES ON TESTS

This clause of IS 302-1 (1979) is applicable except as follows.

4.2 Addition

Tests on separate motor-compressors are covered under a separate Indian Standard.

4.3 Replacement

Tests of 11.7, 9, 10, 11.9 to 11.103 are carried out in this order after tests of 8.

Before testing, the appliance shall be operated at rated voltage for 24 h.

4.5 Addition

Tests according to 9, 10, 11 and 12 are carried out at an ambient temperature of $(32 \pm 1)^{\circ}\text{C}$ on appliances of extended temperate (SN) and temperate (N) temperature class or $(43 \pm 1)^{\circ}\text{C}$ on appliances of subtropical (ST) or tropical (T) temperature class.

Other tests are carried out at an ambient temperature of $(20 \pm 5)^{\circ}\text{C}$.

Appliances rated for several climatic classes are tested at the ambient temperature relevant to the highest climatic class.

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4.6 Addition

Tests are carried out with each source of energy (electricity, gas or fuel) in turn. Tests are additionally carried out with all combinations of sources supplied simultaneously unless this is prevented by interlocking devices.

For the purpose of tests, the optional electric supply ($< 42\text{ V}$) is considered to be extra-low voltage supply.

4.7 Addition

Appliances incorporating ice-makers are tested with the ice-maker operating to give the most unfavourable results.

4.13 Not Applicable

Additional Subclause

4.101 Appliances which are so designed that an ice-maker may be incorporated in it are tested with the intended ice-makers.

5 RATING

The clause of IS 302-1 (1979) is applicable.

6 CLASSIFICATION

The clause of IS 302-1 (1979) is applicable except as follows.

6.1 Addition

According to room temperature:

- Appliances of extended temperature class (SN);
- Appliances of temperate class (N);
- Appliances of subtropical class (ST); and
- Appliances of tropical class (T).

7 MARKING

This clause of IS 302-1 (1979) is applicable except as follows.

7.1 Addition

Moreover, appliances shall be marked with:

- The rated input of heating elements and any auxiliary components if greater than 100 W;
- The defrosting input, in watts, if greater than the input corresponding to the rated input;
- The letters SN, N, ST or T indicating the climatic class of the appliance;
- The rated maximum input of lamps, in watts;
- The chemical name or formula or the refrigerant number of the refrigerant or,

when a mixture is used, the individual name, numbers of chemical formulae.

For compression-type refrigerating systems, the appliance shall also be marked with:

- the mass of the refrigerant, or of each refrigerant in a mixture other than those of the azeotropic type.

Moreover, appliances designed for incorporating an ice-maker shall be marked with the maximum input for an incorporated ice-maker, if greater than 100 W.

Moreover, ice-makers shall be marked with the following:

- The maximum permissible water level, for appliances without automatic water-level control;
- The maximum permissible inlet pressure, in pascals, bars or newtons per cm^2 , for appliances intended to be connected to the water supply mains, unless this is indicated in the instruction sheet; and
- The minimum permissible inlet water pressure in pascals, bars or newtons per cm^2 , if necessary for the correct operation of the ice-maker, unless this is indicated in the instruction sheet.

Furthermore, appliances for camping or similar use shall be marked in addition with the source of supply other than electrical, if any.

Moreover, the means provided for connection of the 12 V supply shall be clearly and permanently marked with the voltage and type of supply for which it is intended.

7.2 Not Applicable

7.12 Addition

For appliances designed for incorporating ice-makers, the instruction sheet shall include the type(s) of ice-maker(s) which can be incorporated.

The instruction sheet shall also include information on the fact that the installation of an ice-maker in an appliance must be made by the manufacturer's services.

According to the type of ice-maker, the instruction sheet shall also include the substance of one of the following warnings:

- Connected to potable water supply only; or
- Fill with potable water only.

Refrigerators and freezers for camping or similar use shall be accompanied with an instruction sheet including the details of the precaution to be taken for safe installation.

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Moreover, the instruction sheet shall include in substance the indication for 'camping use' and if appropriate, the indication of the fact that the appliance is provided with more than one separate supply.

Moreover, the instruction sheet shall include a warning stating that the appliance shall not be used in open air and shall not expose to rain.

7.14 Addition

In the case of free-standing appliances, the marking shall be visible, if necessary after moving the appliance away from the wall.

The marking of the input of the lamp shall be easily discernible while the lamp is being replaced.

7.15 The use of the Standard Mark is governed by the provisions of Bureau of Indian Standards Act, 1986 and the Rules and Regulations made thereunder. The details of conditions under which the licence for the use of Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

8 PROTECTION AGAINST ELECTRIC SHOCK

This clause of IS 302-1 (1979) is applicable except as follows.

8.1 Addition

Protecting covers of lamps, except those which do not need to be removed when replacing the bulb, are removed before applying the standard test finger.

8.2 and 8.3 Not applicable.

8.4 Addition

For the purpose of the test, frozen water is regarded as a conducting liquid.

8.7 Not applicable.

9 STARTING OF MOTOR-OPERATED APPLIANCES

This clause of IS 302-1 (1979) is applicable except as follows.

Addition after the Title

The tests of this clause are carried out at least 15 min after those of **11.7**, but not later than 1 h after the latter. The test of **9.1** is not carried out on appliances incorporating no other motors than motor-compression complying with the locked motor test as specified in relevant Indian Standard on motor compressor and condenser fan motors complying with the locked rotor test of Annex AA.

9.1 Replacement

Motors shall start under all normal voltage conditions which may occur in use.

Compliance is checked by the following test:

The appliance is started three times at 0.85 times the rated voltage at an ambient temperature of 32°C or 43°C according to its class (see **4.5**), the doors and lids being closed.

The appliance is kept switched on after each start for a sufficient time to ensure that the motors start properly and that there is adequate lubrication.

The interval between two successive starts shall be sufficiently long to prevent excessive over-heating of the motor, and to avoid an abnormal increase in the pressure of the liquid refrigerant, as well as to reach equalization of pressure between the high-pressure and low-pressure sides.

Three chatterings of the starting relay of the motor-compressor are permitted.

The supply source is such that the drop in voltage does not exceed 1 percent during the test.

9.2 Replacement

Replace the test specifications after the table by the following:

The wire has a silver content not less than 99.9 percent and is stretched horizontally along the centre line of a box with inside dimensions of 80 mm × 80 mm × 150 mm.

The appliance is then started ten times at a voltage equal to 0.9 times rated voltage and ten times at a voltage equal to 1.1 times rated voltage.

The appliance is kept switched on after each start for a sufficient time to ensure that the motors start properly and there is adequate lubrication.

The interval between two successive starts shall be sufficiently long to prevent excessive overheating of the motors and to avoid an abnormal increase in the pressure of the liquid refrigerant, as well as to reach equalization of pressure between the high-pressure and low-pressure sides.

The chatterings of the starting relay of the motor-compressor are permitted.

During the test, the ambient temperature is 32°C or 43°C according to the class of the appliance (see **4.5**), the doors and lids being closed. Heating elements incorporated in the appliance are operated, but are not connected to a separate supply.

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During the test, the silver wire shall not be melt and neither shall any overload protection device operate.

10 INPUT AND CURRENT

This clause of IS 302-2 (1979) is applicable except as follows.

10.1 Replacement

Replace the test specifications by the following:

Compliance is checked by measuring the input of the appliance under the following conditions:

For appliances other than those pre-set at the manufacturer's works, the thermostat is set to give the lowest temperature. The appliance is empty, with the doors and lids closed and the lamps, if any, switched off.

The tests are carried out at the rated voltage, the appliance being operated at an ambient temperature of 32°C or 43°C according to its class (*see 4.5*).

Ice-makers are operated under normal load.

The result is measured when steady-state conditions are established.

The measured input is the arithmetic mean value of the input between the making and the breaking of the thermostat, or between the highest and the lowest value measured respectively, excluding the starting input but including the input of the incorporated ice-maker, if any.

10.2 Replacement

Replace the test specification by the following:

The test is made under the conditions given in 10.1.

The current is measured when steady-state conditions are established.

The maximum value of the current measured between the making and breaking of the thermostat, excluding the starting current, is considered to be the current taken by the appliance.

Additional Subclauses

10.101 The measured input of a defrosting system shall not exceed the rated defrost input marked on the appliance by more than 10 percent, when such marking is given.

Compliance is checked by measuring the input of the defrosting system when the appliance operates on defrosting, at rated voltage.

10.102 The measured input of any auxiliary component shall not exceed the input marked on the appliance for these components by more than 20 percent, when such marking is given.

Compliance is checked by measuring the input of auxiliary components when operated at rated voltage, when the input is stabilized.

11 TEMPERATURE RISE

This clause of IS 302-1 (1979) is applicable except as follows.

11.2 Replacement

Appliance for building-in are built in, dull black painted plywood walls, about 20 mm thick, being used.

Free-standing appliances are placed in a test enclosure of plywood walls as specified above, the walls enclosing the appliance, as near to all its side and above as possible unless the manufacturer indicates in the instruction sheet that a free distance shall be observed from the walls or the ceiling, in which case this distance is observed during the test.

Other appliances are placed in a test corner. The test corner consists of two walls at right angles, a floor and, if necessary, a ceiling as specified above. The appliance is placed in the test corner as follows:

Appliances normally fixed to a wall are mounted on one of the walls, as near to the other wall and to floor or ceiling of the test corner as is likely to occur in normal use, unless the manufacturer has given other instructions concerning their installation.

Appliances normally fixed to a ceiling are fixed to the ceiling of the test corner as near to the walls as is likely to occur in normal use, provided the manufacturer has not given special instructions concerning their installation.

The appliance is placed in a room maintained at a temperature of 32°C or 43°C according to its class (*see 4.5*) with the doors or lids open until the whole assembly is at room temperature.

11.4 Replacement

Absorption-type appliances and heating elements, if any, are supplied at such a voltage that the input is 1.15 times the maximum rated input.

11.5 Replacement

The resistance of windings of compressor-type appliances is measured before the test when the appliance has reached room temperature.

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The appliance is supplied at the unfavourable voltage between 0.94 times the rated voltage and 1.06 times the rated voltage until steady-state has been reached.

For this test, heating elements are supplied under the conditions indicated in 11.4.

The measurement of the resistance of the windings is not made on motor-compressors complying with relevant Indian Standard.

11.6 Replacement

In general, the supply conditions specified for absorption-type appliances apply also to motors incorporated in such appliances. In case of doubt, however, the conditions specified below apply.

For combined appliances, when the motors are operated at a voltage equal to 1.06 times the maximum rated voltage, the input to the heating elements is as specified in 11.4. When the motors are operated at a voltage equal to 0.94 times the minimum rated voltage, the input to the heating elements is reduced to 0.90 times the minimum rated input.

If it is necessary to make the test as an intermediate voltage, the input of the heating elements is adjusted proportionately.

11.7 Replacement

Except for the tests of 11.10 to 11.103, the thermostat or similar control device is short-circuited and doors or lids are closed.

The appliance is then operated until steady-state conditions are established.

Ice-makers and incorporated ice-makers are operated under normal load.

Temperatures of the different parts of the appliance are then measured.

11.8 Replacement

Replace the text above the table by the following:

During the test and before steady-state conditions are established the overload protection devices, if any, may operate. When steady-state conditions are established these devices shall not operate. During the test, sealing compound, if any, shall not flow out. For the test, the appliances are considered as liable to operate continuously for long periods.

For the appliances of extended temperature or temperate classes, the temperature rises are measured and shall not exceed the value given in the following table.

For appliances of subtropical or tropical classes, the temperature rises are measured and shall not exceed the value given in the following table reduced by 7°C.

Addition after the Table

Moreover, the temperature of windings and the enclosure of the hermetic motor-compressors other than those complying with the Indian Standard are measured and shall not exceed the following values:

<i>Parts of the Appliance</i>	<i>Temperature °C</i>
Winding of the hermetic motor-compressors	
— Synthetic insulation	140
— Cellulosic insulation or the like	130
External enclosures of hermetic motor-compressors	150

11.10 Replacement

If the winding temperature of a hermetic motor-compressor other than those complying with relevant Indian Standard is higher than the temperature limit indicated in the table, the test is carried out again, the thermostat or similar control device being set at the lowest temperature, and the short-circuit of the device removed.

The winding resistance is measured at the end of a running cycle.

The corresponding temperature shall be not higher than the temperature limit indicated in the table.

Additional Subclauses

11.101 Any defrosting assembly shall not give rise to temperatures likely to cause a fire or serious damage to the electrical insulation, combustible material, electro-valves or other electrical components.

Compliance is checked by the following test

The appliance is supplied at the most unfavourable voltage between 0.94 and 1.06 times the rated voltage:

— In the case of appliances where defrosting is manually controlled, until the evaporator is coated with a layer of ice or frost.

A method of accumulation of frost for refrigerators only is given in 11.102. For other appliances, a method is under consideration.

— In the case of appliances where defrosting is automatically or semi-automatically controlled, until the evaporator is coated

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with a layer of frost; however, this layer shall be not thicker than that which occurs, in normal use, during the intervals between the successive automatic defrosting operations, or, for the semi-automatic defrosting during the intervals between the defrosting operation recommended by the manufacturer.

The defrosting assembly is then switched on by closing the circuit supplying the heating element or the valve, the supply voltage of the heating element being as defined in 11.4.

When it is not possible to disconnect the heating element, the test is made with a supply voltage equal to 1.06 times the rated voltage for all the elements.

If the defrosting time is controlled by an adjustable device, the device is set to the time given by the manufacturer. If a control device is used which stops the defrosting at a given evaporation temperature or pressure, the defrosting period is automatically terminated when the control operates.

The temperatures of combustible materials and of electrical components liable to be affected by the defrosting operation are measured with thermocouples.

The test is continued until the minimum temperatures are reacted. The temperatures and temperature rises shall not exceed the values given in the table of 11.8.

During the recovery period after defrosting, the thermal overload protector may operate.

11.102 The accumulation of frost shall be produced by the use of a device having a controllable heat source directed on a measured amount of water for the purpose of evaporating this water over a predetermined period with a minimum of extraneous heat loss to the refrigerator cabinet.

A convenient form of the apparatus would comprise a block enclosure of thermally insulating material having a vertical hole at its centre containing a light bulb mounted on a bottom plug directly below an operating dish with a high thermal conductivity base and low thermal conductivity walls (see Fig. 101A and 101B).

The device described above should be mounted at the geometric centre of the refrigerator cabinet and the electrical connection brought conveniently to the outside so that the voltage applied may be varied and the power consumed measured with the refrigerator door in the closed position.

Water is then introduced into the evaporating dish at the required rate through a length of small bore tube passing into the cabinet. A continuous flow is not necessary but the water should be injected at appropriate intervals.

Provision shall be made (for example, in the control of the supply of electrical energy to the device) to ensure that the evaporation of water under normal conditions of use is capable of being maintained at a rate equal to:

2 g water/dm³ of cabinet gross volume/week.

The electric energy to the device shall not be excessive but shall be sufficient to ensure the complete evaporation of the water.

The amount of frost to be accumulated prior to the start of the defrosting test shall be based on this rate and on the time interval between two successive defrosts according to the manufacturer's instructions.

For example, if the manufacturer's instructions recommend defrosting twice weekly, then a refrigerator with a cabinet gross volume of 140 dm³ will require:

$$2 \text{ g} \times \frac{140}{2} = 140 \text{ g of water.}$$

The rate given above may be exceeded in certain circumstances.

The apparatus described (see Fig. 101A and 101B) has a maximum evaporating rate of approximately 2 g per hour when operating with an input of 4 W and with the water to be evaporated entering at cabinet temperature.

Vision of the method of accumulation of frost is under consideration.

11.103 Ancillary heating element, other than defrosting assemblies incorporated in an appliance shall not give rise to excessive temperatures.

Compliance is checked by the following test:

Ancillary heating elements are energized, with the refrigerating device switched off if this is possible in normal use. They are supplied with a voltage as given in 11.4, until steady state conditions are reached.

Temperature rises are measured by means of thermocouples fixed on the outside surface of the insulation of the ancillary heating element. Temperature rises shall not exceed the values specified in 11.8.

12 OPERATION UNDER OVERLOAD CONDITIONS OF APPLIANCES WITH HEATING ELEMENTS

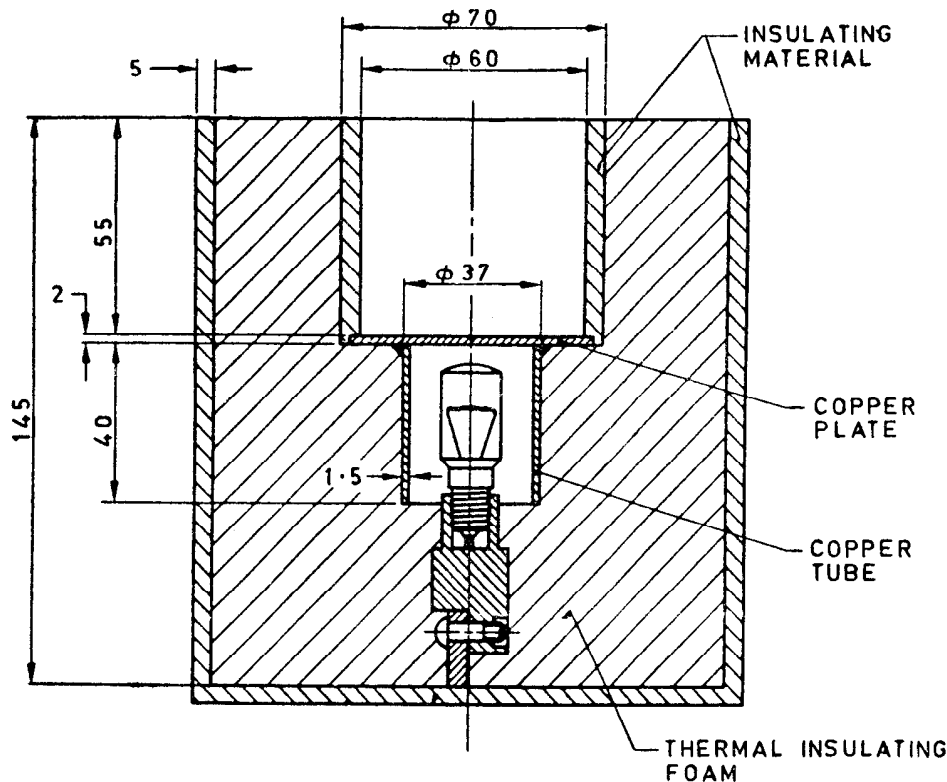
12.1 This clause of IS 302-1 (1979) is applicable except as follows.

12.1 Replacement

Replace the test specifications by the following:

For absorption-type appliances, compliance is checked by the tests of 12.2 carried out at an ambient temperature of 32°C or 43°C according to the class of the appliance (see 4.5).

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All dimensions in millimetres.

FIG. 101A WATER EVAPORATION FOR ACCUMULATION OF TEST

Ancillary heating elements are not tested since they are subjected to 19.101.

13 ELECTRICAL INSULATION AND LEAKAGE CURRENT AT OPERATING TEMPERATURE

13.1 Replacement

Replace the test specifications by the following:

Compliance is checked by the tests of 13.2 and, for absorption-type appliances and other heating elements by the additional test of 13.3, the appliance being operated under the conditions specified in 11.7.

14 RADIO AND TELEVISION INTERFERENCE SUPPRESSION

This clause of IS 302-1 (1979) is applicable.

15 MOISTURE RESISTANCE

This clause of IS 302-1 (1979) is applicable except as follows.

15.1 Replacement

Appliances shall be proof against splashing or spilling of liquids and humid conditions which may occur in normal use.

Compliance is checked by the following tests:

— Appliances other than ordinary are tested by the appropriate test of 15.2;

Appliances subject to spillage of liquid in normal use are tested by the tests of 15.3;

All appliances are tested by the tests of 15.4;

— Appliances subject to spilling of liquid inside are tested by the tests of 15.101;

Appliances except those for building-in and ice-makers by the tests of 15.102; and

Ice-makers which are directly connected to the water supply by the tests of 15.103.

NOTE — If there is a possibility that water deposit during a test could affect the results of the following test, the appliances is kept in normal test room atmosphere for 24 h before being subjected to the next test.

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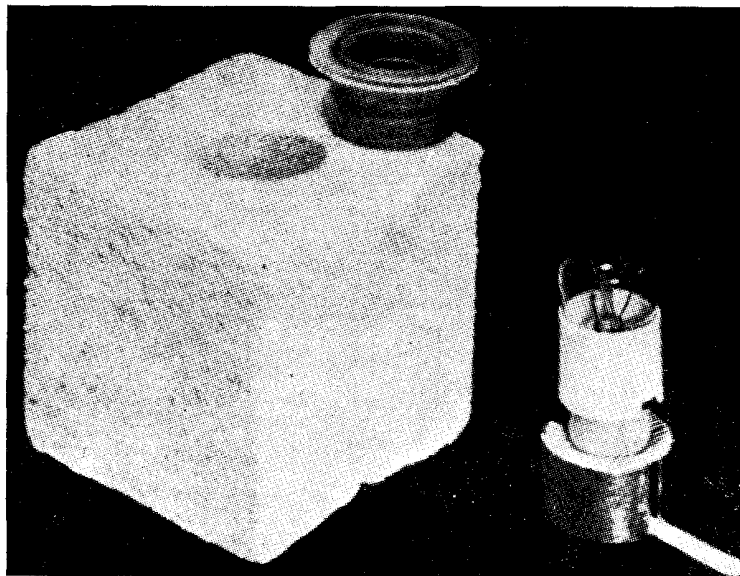
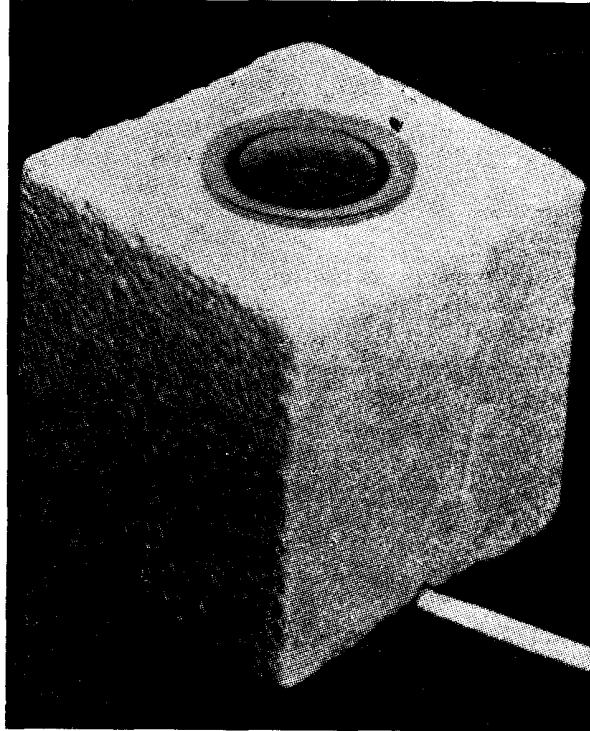


FIG. 101 B WATER EVAPORATION DEVICE FOR ACCUMULATION OF FROST

After each of these tests, except **15.4**, the appliance shall withstand the electric strength test specified in **16.4** and inspection shall show that water which may have entered the appliance, does not impair compliance with this standard; in particular, there shall be no trace of water on insulation which could result in the reduction of creepage distances and clearances below the limits specified in **29.1**.

Immediately after the test of **15.4**, the appliance shall withstand the tests of **16**.

15.2 Addition (beginning of the subclause)

The enclosure of drip proof, splash proof and watertight appliances shall provide the degree of protection against moisture in accordance with the classification of the appliance.

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15.3 Replacement

The fourth paragraph of the test specifications:

‘Immediately after this treatment sub-clause 29.1’ is not applicable.

15.4 Replacement

Replace the first two paragraphs to the test specifications by the following:

Compliance is checked by the humidity treatment described in this subclause followed immediately by the tests of 16.

The appliance is placed with its doors or lids open in the humidity cabinet and does not operate.

Additional Subclauses

15.101 Appliances subject to spillage of liquid on the inside walls of the cabinet of compartment shall be so constructed that such spillage does not affect their electrical insulation.

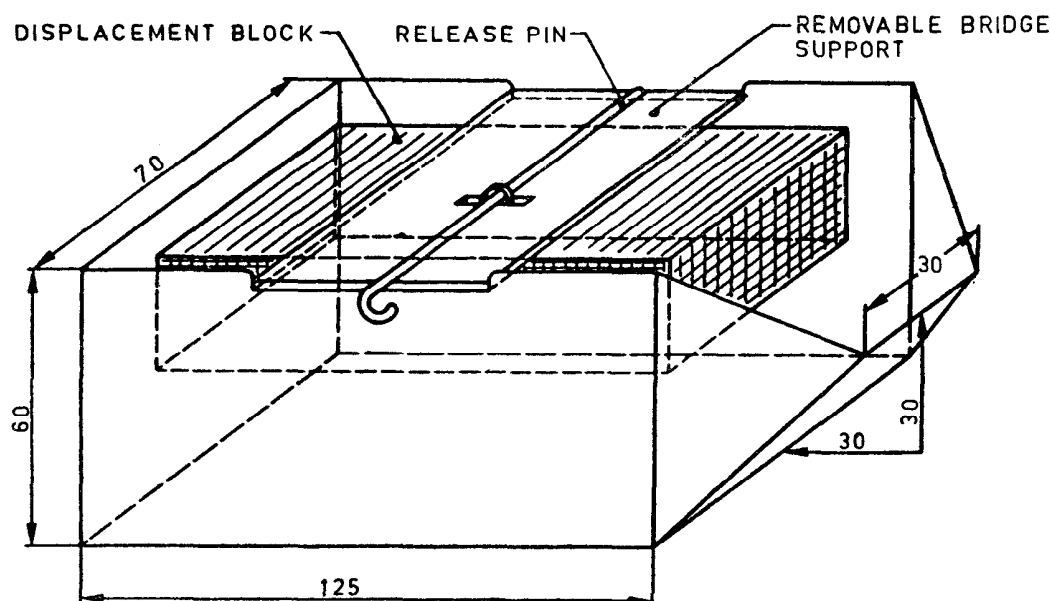
All shelves and containers which can be removed without the use of a tool are removed and the inside walls of the cabinet or compartment including any electrical components mounted thereon are subjected to the following tests,

the appliance being disconnected from the supply.

The apparatus shown in Fig. 102 is filled with water to the level of the lip and the displacement block is supported just above the water by means of any suitable release mechanism and bridge support.

The apparatus is supported with its base horizontal and so positioned and at such a height that the water is discharged over the part under consideration within the interior of the appliance and its front face in the most unfavourable manner when the release mechanism is operated. The test is made only once with the apparatus in any one position but the test may be repeated as many times as necessary in different positions provided that there is no residual water on parts wetted by a previous test.

15.102 Appliances other than those for building-in only and other than ice-makers are tilted at an angle of up to 2° to the position of normal use in the direction which is likely to be the most unfavourable for this test, and half a litre of water containing approximately one percent NaCl is poured uniformly over the top of the appliance with the controls in the ‘on’ position,



All dimensions in millimetres.

The displacement block has a volume of 140 ± 5 ml and a mass of 200 ± 10 g, dimensions are approximately $112 \times 50 \times 25$.

The dimensions of the vessel are inside dimensions and the tolerance is ± 2 .

FIG. 102 APPARATUS FOR SPILLAGE TEST

the appliance being disconnected from the supply.

15.103 For ice-makers which are directly connected to the water supply, the container and/or that part of the appliance which serves as container is filled with water as in normal use and the inlet valve is left open until spillage occurs and additional for a further minute.

Where no spillage occurs owing to the fact that advice prevents such spillage, the inlet valve is rendered in operative for a further 5 min following the operation of this device.

16 INSULATION RESISTANCE AND ELECTRIC STRENGTH (After the Humidity Treatment)

This clause of IS 302-1 (1979) is applicable, except as follows.

16.3 Addition

Add the following item in the Table:

For appliances for camping or similar use.

Between circuits supplied at extra-low voltage and the mains supply 7 M

16.4 Addition

Add the following item in the Table:

	<i>Class III Appli- ances</i>	<i>Class II Appli- ances</i>	<i>Other Appli- ances</i>
For appliances for camping on similar use between circuits supported at extra-low voltage and the main supply	—	3 750 V	3 750 V

17 OVERLOAD PROTECTION

This clause of IS 302-1 (1979) is applicable.

18 ENDURANCE

This clause of IS 302-1 (1979) is not applicable.

19 ABNORMAL OPERATION

This clause of IS 302-1 (1979) is applicable except as follows.

19.1 Replacement

Appliances shall be so designed that the risk of fire, mechanical damage related to safety of electric stock as a result of abnormal or careless operation is obviated as far as is practicable.

Compliance is checked:

- For compressor-type applications, by the test carried out on a separate motor-

compressor, only once for a given type of motor-compressor in the conditions stated in relevant Indian Standard, unless the motor-compressor has been previously tested and complies with the relevant Indian Standard;

- For absorption-type appliances, by the tests of 12;
- Furthermore, for appliances provided with a condenser fan motor, by the test of 19.6 and the locked-rotor test carried out on a separate motor, only once for a given type of fan motor under the conditions stated in Annex AA;
- Temperatures of motor windings other than condenser fan motors are not measured;
- Winding temperatures of motor-compressors are not measured;
- Furthermore, for appliances provided with other motors, by the test of 19.6;
- Furthermore, for ice-makers, by the test of 19.102; and
- Furthermore, for appliances for camping or similar use, by the test of 19.103.

19.2 to 19.5 Not applicable.

19.7 Replacement

Appliances incorporating three-phase motors, except three-phase motor-compressors complying with relevant Indian Standard, are operated under normal load, with one phase disconnected for a period as specified in 19.6.

Further details of this test are under consideration.

19.8 Not applicable.

19.9 Replacement

Illuminating equipment shall not cause any life hazard under abnormal operating conditions.

Compliance is checked by the following tests:

The complete illuminating equipment including its protective cover, fitting with a lamp as recommended by the manufacturer, is operated for 12 h at 1.06 times rated voltage. The refrigerator unit is switched off, the appliance is empty and doors or lids are fully opened.

If the lamp does not attain the rated maximum input at rated voltage, the voltage is varied until the rated input is reached and then increased to 10.6 times this voltage.

During the test, no development of smoke

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shall appear and surrounding plastic parts shall not show any distortion which may affect in the sense of this standard.

After the test, protection against electric shock shall be in compliance with **8** and the electric strength test according to **16** shall be withstood, the test voltage being applied only between current-carrying parts and accessible metal parts.

19.11 Addition

The temperature of the enclosure of motor-compressors other than those which have been tested separately according to the relevant Indian Standard, is determined at the end of the test period or at the instant of operation of fuses, thermal cut-outs, motor protection devices and the like, and shall not exceed 150°C.

Additional Subclauses

19.101 Ancillary heating elements shall be so dimensioned and located that there is no risk of fire even in the case of abnormal operation.

Compliance is checked by inspection and the following test:

Ancillary heating elements are operated continuously at a voltage equal to 1.1 times rated voltage of the appliance, under steady conditions are established. If there is more than one ancillary heating element, they are operated each in turn, unless failure of a single component will cause two or more to operate together in which case they are tested in combination. If such a heating element is intended to be switched on and off by the use, it is operated together with the above combination.

If a thermal cut-out operates or if the current is otherwise interested, for example, due to the rupture of an intentionally weak part before steady-state conditions are established, the heating period is considered to be terminated.

During this test, there shall be no emission of this smoke, flames or molten metal.

The refrigerating unit is not switched off if this prevents the heating element to operate.

19.102 Ice-makers shall be so designed that the risk of life, mechanical hazard or electric shock is obviated as far as it is practicable in the event of incorrect operation of the development of defects in control devices or in their associated devices.

Compliance is checked by applying any defect which may be expected in normal use, while the ice-maker and incorporated ice-maker is operated under normal load and at rated voltage or

at the upper limit of the rated voltage range. Only one fault condition is reproduced at a time, the tests being made consecutively.

During the tests, the ice-maker or the appliance incorporating an ice-maker shall not emit flames or motion metal and the temperature of windings shall not exceed the values shown in the table of **19.6**.

After the test, the appliance shall comply with the requirements of **19.11**.

Examples of fault conditions are:

- Timber stopping in any position;
- Disconnection and reconnection of one or more phases of the supply during any part of the programme;
- Open-circuiting or short-circuiting of components;
- Failure of magnetic valve; and
- Operation with an empty container.

In general, tests are limited to those cases which may be expected to give the most unfavourable results.

The tests of this subclause are made with the tap closed or opened whichever gives the most unfavourable results.

For the purpose of these tests, thermal controls are not short-circuited.

Components complying with the relevant Indian Standard are not open-circuited or short-circuited, provided the appropriate standard covers the conditions which occur in the appliance.

Water level switches complying with IEC 1058 are not short-circuited during these tests.

The test during which the automatic filling device is held open has already been made during the test of **15.3**.

19.103 For appliances intended for camping or similar use compliance is checked by operating the appliances at an ambient temperature of 32°C or 43°C according to the class of the appliance. The appliance is placed on a support inclined in the most unfavourable position by 5° and is operated at rated voltage or rated data of other source of energy respectively until steady-state conditions are established.

During the test, non-self-resisting thermal cut-outs which are only accessible with the aid of a tool or which require the replacement of a part shall not operate and no vapour of ignitable gas shall accumulate in the appliance.

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20 STABILITY AND MECHANICAL HAZARDS

This clause of IS 302-1 (1979) is applicable except as follows.

20.1 This subclause is applicable to ice-makers only.

Moreover for refrigerators and freezers, the test of Annex BB applies.

21 MECHANICAL STRENGTH

This clause of IS 302-1 (1979) is applicable except as follows.

Additional Subclause

21.101 Appliances for camping or similar use, other than fixed appliances and appliances for building in shall withstand the following tests:

The appliance is placed on a horizontal wooden panel which is dropped 50 times from a height of 5 cm onto a solid base or wood.

The appliance is then fastened in its normal position of use to the vibration-generator by means of straps round the enclosure. The direction of vibration is vertical, and the severity is:

— Duration	30 min
— Amplitude	0.35 mm
— Sweep frequency range	10 Hz, 55 Hz, 10 Hz
— Sweep rate	Approximately one octave per minute

After the tests, the appliance shall show no damage affecting the safety; in particular, no connection or part the loosening of which might impair safety shall have loosened.

22 CONSTRUCTION

This clause of IS 302-1 (1979) is applicable except as follows.

22.1 Replacement

Appliances shall not be of Class 0 or Class 01.

Compliance is checked by inspection and by the relevant test.

22.3 Modification

Replace the test specification by the following:

Compliance is checked by verifying that the appliance operations without dangers in all positions from the normal position of use by an angle not exceeding 2°.

22.8 and 22.9 Not applicable.

22.11 Addition

Thermostats, with the exception of their thermo-sensitive basis, shall not be arranged inside the evaporator unless they are adequately protected against condensation on cold surface and against the effect of water formed during the defrosting process.

Compliance is checked by inspection part such as storms and tube of thermostat.

22.15 Not applicable.

22.26 and 22.29 Not applicable.

Additional Subclause

22.101 Lamps shall be effectively protected against mechanical shocks either by devices placed on their support or by the supports themselves.

Lampholders shall be so fixed that they do not work loose in normal use.

Normal use includes replacement of the bulb.

Compliance is checked by inspection and, if necessary, by a test subjecting the lampholders to a torque of 0.15 Nm for E14 and B15 lampholders, and 0.25 Nm for E27 and B22 lampholders. The lampholders shall then withstand a pull test with 50 N, applied for 1 min in the direction of the axis of the lampholders.

22.102 Mains-operated heating conducts (defrost heaters, door heaters, butter heaters, etc) and their joints, located in the refrigerator/freezer compartments or in the thermal insulation shall be watertight.

Compliance is checked by immersing three samples of the complete heating conductor in water containing approximately 1 percent NaCl and having a temperature of $(27 \pm 5)^\circ\text{C}$ for a period of 24 h.

A voltage of 2 000 V is then applied for 15 min between the heating conductor(s) and the water.

During the test, no breakdown shall occur.

In the meaning of the paragraph, connections to electrical terminals are not joints.

22.103 Ice-makers and appliances incorporating ice-maker shall withstand the water pressure to which may be subjected in normal use.

Compliance is checked by subjecting those parts of the ice-makers and appliances incorporating ice-maker which are under pressure from the water supply mains, for 5 min to a static pressure equal to twice the maximum permissible inter water pressure or 1.2 MPa = 12 bar = 120 N/cm², whichever is the greater.

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During the test, there shall be no leakage from any part including the inter water hose.

23 INTERNAL WIRING

This clause of IS 302-1 (1979) is applicable.

24 COMPONENTS

This clause of IS 302-1 (1979) is applicable except as follows.

24.1 Addition

For stating capacitors, the voltage across the capacitors shall not exceed 1.3 times the rated voltage of the capacitor when the appliance is operating at 1.1 times the rated voltage.

Motor-compressors are not required to be separately tested according to relevant Indian Standard nor as they required to meet all requirements of relevant Indian Standards if they meet all requirements of the appropriate appliance standard.

24.3 Addition

Voltage selection switches used in appliances for similar use shall disconnect all poles and shall have a contact separation of at least 3 mm.

24.4 Not applicable.

24.5 Replacement

Socket-outlets inside Class 1 appliances are permitted; they shall be provided with an earth-contact.

Compliance is checked by inspection and by a manual test.

24.6 Addition

Lampholders shall be of the insulated type.

Compliance is checked by inspection.

24.9 Not applicable.

25 SUPPLY CONNECTION AND EXTERNAL FLEXIBLE CABLES AND CORD

This clause of IS 302-1 (1979) is applicable except as follows.

This clause of IS 302-1 (1979) is not applicable to motor-compressors complying with the relevant Indian Standard.

25.1 Replacement

Replace the requirement by the following:

Appliances shall be provided with one of the following means of connection to the supply:

- A set of terminals for permanent connection to fixed wiring;

- A set of supply leads for permanent connection to fixed wiring;
- An appliance inlet; and
- A power supply cord.

Appliances shall not be provided with more than one means of connection to the supply, with the exception that appliances consisting of two or more completely indented unit; built together in one casing may be provided with more than one set of terminals or with more than one power supply cord.

Addition

Appliances for camping or similar use intended for optional connection to the mains supply and to a 12 V supply shall be equipped with a mains supply cord and a supply cord or separate terminals for the connection of the 12 V supply.

25.3 Addition

It shall not be possible to remove the connector from the appliance by a pull on the flexible cord before an independent operation to release the locking means has been carried out.

25.6 Replacement

The light polyvinyl chloride sheathed flexible cord complying with IS 694 : 1977 is also allowed for appliances having a mass exceeding 3 kg.

Addition

For rated currents up to 3 A, a nominal cross-sectional area of 0.5 mm² is allowed provided the length of the cord does not exceed 2 m.

The length of the cord is measured between the point where the cord, or cord guard enters the appliance or connector, and the entry to the plug.

26 TERMINALS FOR EXTERNAL CONDUCTORS

This clause of IS 302-1 (1979) is applicable, except for parts related to motor-compressors, if the motor-compressor complies with relevant Indian Standard.

27 PROVISION OF EARTHING

This clause of IS 302-1 (1979) is applicable, except for part related to motor-compressors if the motor-compressor complies with relevant Indian Standard.

28 SCREWS AND CONNECTIONS

This clause of IS 302-1 (1979) is applicable, except for parts related to motor-compressors if the motor-compressor complies with relevant Indian Standard.

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29 CREEPAGE DISTANCES, CLEARANCES AND DISTANCES THROUGH INSULATION

This clause of IS 302-1 (1979) is applicable, except for parts related to motor-compressors if the motor-compressor complies with IEC 335-2-34.

29.1 Addition

Add to the table the following:

In appliance for camping or similar use:

'between deperated circuits
for optional operation
with extra-low voltage
and main supply' [—|—|8|8|8|—|—|]

30 RESISTANCE TO HEAT, FIRE AND TRACKING

This class of IS 302-1 (1979) is applicable, except for parts related to motor-compressors if the motor-compressor complies with relevant Indian Standard.

30.1 Addition

Accessible parts of non-metallic material within the storage compartment are regarded as external parts.

For accessible parts of non-metallic material within the storage compartment, the temperature value of $(75 \pm 2)^{\circ}\text{C}$ is replaced by $(65 \pm 2)^{\circ}\text{C}$.

31 RESISTANCE TO RUSTING

This clause of IS 302-1 (1979) is applicable, except for parts related to motor-compressors if the motor-compressor complies with the relevant Indian Standard.

32 RADIATION, TOXICITY AND SIMILAR HAZARDS

This clause of IS 302-1 (1979) is not applicable.

101 TESTS

101.1 Type Test

The tests specified in Table 101 shall constitute the type tests and shall be carried out on two samples of the same type and rating selected preferably at random from a regular production lot. Before commencement of the tests, the samples shall be visually examined and inspected for obvious visual defects in respect of components, parts of their assembly, construction, mechanical hazards, markings, provision of suitable terminals for supply connections, earthing and the effectiveness of screws and connections. The external surface finish shall be even and free from finishing defects.

Table 101 Schedule of Type Tests
(Clause 101.1)

Sl No.		Clause Reference
i)	Protection against electric shock	8
ii)	Starting of motor operated appliances	9
iii)	Input	10
iv)	Temperature rise	11
v)	Operation under overload conditions	12
vi)	Electrical insulation and leakage current at operating temperature	13
vii)	Radio and television interference suppression	14
viii)	Moisture resistance	15
ix)	Insulation resistance and electric strength (after humidity treatment)	16
x)	Overload protection	17
xi)	Endurance	18
xii)	Abnormal operation	19
xiii)	Stability and mechanical hazards	20
xiv)	Mechanical strength	21
xv)	Construction	22
xvi)	Internal wiring	23
xvii)	Components	24
xviii)	Supply connection and external flexible cables and cords	25
xix)	Terminals for external conductors	26
xx)	Provision for earthing	27
xxi)	Screws and connections	28
xxii)	Creepage distance and clearances	29
xxiii)	Resistance to heat, fire and tracking	30
xxiv)	Resistance to rusting	31
xxv)	Radiation hazards	32

101.1.1 Criteria of Acceptance

Both samples shall successfully pass all the type tests for proving conformity with the requirements of the standard. If any of the samples fails in any of the type tests, the testing authority at its discretion, may call for fresh samples not exceeding twice the original number and subject them again to all tests or to the test(s) in which failure(s) had occurred. No failure should be permitted in the repeat test(s).

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101.2 Acceptance Tests

The following shall constitute the acceptance tests:

<i>Test</i>	<i>Clause Reference</i>
a) Protection against electric shock	8
b) Input	10
c) Temperature rise	11
d) Electrical insulation and leakage current at operating temperature	13
e) Moisture resistance	15
f) Insulation resistance and electric strength (after humidity treatment)	16

g) Provision for earthing 27

NOTE — For the purpose of acceptance tests, the humidity treatment shall be done for 24 hours while conducting the test for moisture resistance (15).

101.2.1 A recommended sampling procedure for acceptance tests is given in Appendix B of IS 302-1 (1979).

101.3 Routine Test

The following shall constitute the routine tests:

<i>Test</i>	<i>Clause Reference</i>
a) Protection against electric shock	8
b) High voltage	13.3.2 of IS 302-1 (1979)
c) Provision for earthing	27

ANNEX A

TABLES OF TYPE TESTS

Appendix A of IS 302-1 (1979) is not applicable.

ANNEX B

SAMPLING PROCEDURE FOR ACCEPTANCE TESTS

Appendix B of IS 302-1 (1979) is not applicable.

ANNEX C

ELECTRONIC CIRCUITS

Appendix C of IS 302-1 (1979) is applicable if electronic circuits are used.

ANNEX D

MEASUREMENT OF TEMPERATURE WITH THERMOMETER

Appendix D of IS 302-1 (1979) is applicable.

ANNEX E

ALTERNATIVE TESTS FOR PROTECTED MOTOR UNITS

Appendix E of IS 302-1 (1979) is not applicable to motor-compressors and to condenser fan motors.

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ANNEX F
IMPACT TESTS APPARATUS

Appendix F of IS 302-1 (1979) is applicable.

ANNEX G
THERMAL CONTROLS AND OVERLOAD RELEASES

Appendix G of IS 302-1 (1979) is applicable if thermal controls and overload releases are used.

ANNEX H
MEASUREMENT OF CREEPAGE DISTANCES AND CLEARANCES

Appendix H of IS 302-1 (1979) is applicable.

ANNEX J
TESTS FOR FIRE RESISTING PROPERTIES

Appendix J of IS 302-1 (1979) is applicable.

ANNEX K
**BNF JET TEST FOR DETERMINATION OF THICKNESS OF COPPER
AND NICKEL PLATING**

Appendix K of IS 302-1 (1979) is not applicable.

ANNEX L
**APPROXIMATE MEASUREMENT OF THICKNESS OF CHROMIUM ON
NICKEL, STEEL AND COPPER**

Appendix L of IS 302-1 (1979) is not applicable.

ANNEX AA
(Clause 9)

LOCKED-ROTOR TEST OF CONDENSER FAN MOTORS

The enclosure and winding of a condenser fan motor shall not reach excessive temperatures if the motor locks or fails to start.

Compliance is checked by the following test:

The fan and its motor are mounted on wood or similar material. The motor's rotor is locked. Fan blades and motor brackets are not removed.

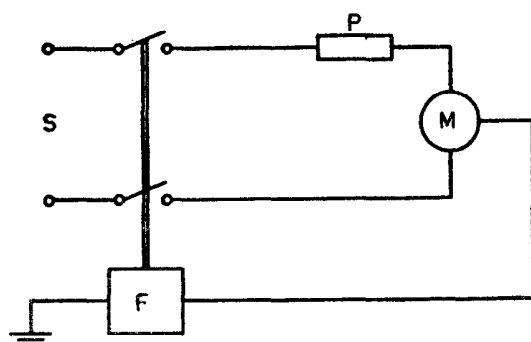
The motor is supplied at rated voltage; the supply circuit is given in Fig. AA1.

The assembly is to operate under these conditions for 15 days (360 h), unless the protective device, if any, permanently open-circuits by failure period to the expiration of that time. In this case, the test is discontinued.

If the temperature of motor windings stays lower than 90°C, the test is discontinued when steady-state conditions are reached.

Temperatures are measured under conditions stated in 11.3.

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S = Supply
M = Fan motor
P = Protection device (external or internal)
F = Leakage circuit breaker 30 mA

FIG. AA1 SUPPLY CIRCUIT FOR LOCKED-ROTOR
TEST OF CONDENSER FAN MOTOR

During the test, enclosure and winding temperatures shall not exceed the values stated in 19.6.

The polarity of the supply shall be inverted every 24 h.

Seventy-two hours after the beginning of the test, the motor shall withstand an electric strength test as specified in 16.4.

An earth-leakage circuit-breaker is connected so as to disconnect the supply in the event of an earth current of more than 30 mA.

At the end of the test, the leakage current is measured between windings and the body at a voltage equal to twice the rated voltage; its value shall not exceed 2 mA.

ANNEX BB

(*Clauses 1.1 and 20.1*)

STABILITY AND MECHANICAL HAZARDS

Appliances including ice-makers intended to be used on a surface such as a floor or a table shall have adequate stability.

Compliance is checked by the following tests:

The empty appliance is disconnected from the supply and placed on a horizontal support and the door is opened through an angle of approximately 90°. Cylindrical weights, having a diameter of 80 mm and a mass of 0.5 kg, are placed on each of the door shelves starting as far as possible from the hinge and touching each other all along the shelf, except for a space less than 8 cm wide at the end of the shelf. Two of these weights are placed in each position on these shelves where a milk bottle can be accommodated, whereas one weight only is placed on all other shelves, excepting those which are specifically designed for storing eggs only.

A weight having a mass of 2.3 kg is then placed on top of the door, 4 cm from the edge furthest from the hinge, the door being opened through an angle of approximately 90°.

This test is repeated with the door opened through an angle of approximately 180° or to the limit of the door-stop, whichever results in the smaller angle.

During these tests, the appliance shall not tilt.

For appliances provided with more than one door, the tests are made on each door separately.

It is not necessary for the weights to lie flat on the shelves. If the shelf is too narrow, the weights may either overhang or be tipped up, but there shall be no more than one weight, or two weights for milk bottles, for each 8 cm length of the shelf.

If egg racks can be removed, the relevant shelf is not considered to be specifically designed for storing eggs only. Damage to, and deformation of, doors, hinges and shelves are neglected.

In the case of appliances with a volume of 60 litres or more, and for reasons of safety, the door, if it is not locked, should be capable of being opened from the inside.

In addition, the force required for opening the door shall not exceed 70 N.

Compliance is checked by the following test:

The door shall be maintained closed for a period of 1 h after which an 'opening test' shall be carried out under the following condition:

The force of opening shall be applied to the handle at the point farthest from the hinges, in a direction perpendicular to the plane of the door, by means of a spring balance.

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ANNEX CC

(Foreword)

The following Indian Standards are necessary adjuncts to this standard:

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
		694 : 1990	PVC insulated cables for working voltages up to and including 1 100 volts (<i>third revision</i>)
302-1 (1979)	Safety of household and similar electrical appliances: Part 1 General requirements (<i>fifth revision</i>)	9968 (Part 1) : 1988	Elastomer insulated cables: Part 1 For working voltages up to and including 1 100 volts (<i>first revision</i>)

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